

Statement of Work
Medical Physicist
Services
Rosebud Service Unit

Section 1: General Information

1.1 General:

The Mission of the Rosebud Service Unit is to provide the best possible healthcare services to the patients on the Rosebud Indian Reservation. To meet this mission, the Rosebud Service Unit is requesting a non-personal services contract for a Medical Physicist. The Government shall not exercise any supervision or control over the contract service providers performing the services herein. Such contract service providers shall be accountable solely to the Contractor who, in turn is responsible to the Government.

1.2 Period of Performance: Will be one (1) year from the Date of Award

1.3 Physical surveys must be conducted on an annual basis 12 months following the previous year's survey.
All reports are due within 30 days of completed survey. Assistance to create corrective action plan, review plan with program director, and re-evaluate to ensure compliance is acceptable may occur within the following 11 months. Total Period of performance is March 1, 2026 through February 28, 2027.

1.4 Place of Performance: Rosebud Service Unit
 PO Box 550
 800 Hospital Drive
 Rosebud, SD 57570

1.5 Hours of Operation: All work shall be performed during business hours of 8:00 am to 5:00 PM. Hours can differ if the facility is open for business 24/7 or if mutually agreed between the facility and contractor representative for the benefit of the surveyor. If a service has begun during regular business hours it must be completed before any patients can be imaged according to the American College of Radiology (ACR), The Joint Commission (TJC) and the Mammography Quality Standards Act (MQSA).

1.6 Type of Contract: The government will award a Firm Fixed Price Contract

1.7 Invoicing: All invoices from the contractor shall be submitted electronically via the IPP (Invoice Payment Platform) which routes to the COR for Payment Processing.

1.8 COR Information
Jessie Wiedman, COR I Jessie.Wiedman@ihs.gov

1.9 Contractor Authority

1. Authority

1.1. No person other than the Contracting Officer has authority to bind the Government with respect to this contract.

1.2. No action or omission of any government employee or representative other than the Contracting Officer shall increase or decrease the scope of this contract or shall otherwise modify the terms and conditions of this contract.

1.3. In no event shall any of the following be effective or binding on the Government or imputed to the Contracting Officer with respect to this contract:

1.3.1. An understanding or agreement between the Contractor and anyone other than the Contracting Officer.

1.3.2. A purported modification or change order issued by anyone other than the Contracting Officer.

1.3.3. A promise by anyone other than the Contracting Officer to provide additional funding or make payments; or

1.3.4. An order, direction, consent, or permission from anyone other than the Contracting Officer to:

1.3.4.1. Incur costs in excess of a specified estimated cost, allotment of funds, or other ceiling; or

1.3.4.2. Expend hours in excess of a specified level of effort.

Section 2: Definitions & Acronyms

2.1 Definitions:

Contractor. A supplier or vendor awarded a contract to provide specific supplies or service to the government.

Work Day. The number of hours per day the Contractor provides services in accordance with the contract is an 8-hour day.

2.2 Acronyms:

AAPM
ACR
CBCT
CFR
COR-NA
CT
FOV
HVL
IEC
MQSA
MRI
NCRP
PET
QA
RSO
SMPTE
SOW

SPECT TJC

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lege of Radiology
Cone Beam Computed Tomography
Code of Federal Regulations
Contracting Officer Representative-Non-Applicable
Computed Tomography
Field of View
Half Value Layer
International Electrotechnical Commission
Mammography Quality Standards Act
Magnetic Resonance Imaging
National Council for Radiation Protection
Positron Emission Tomography
Quality Assurance
Radiation Safety Officer
Society of Motion Picture and Television Engineers
Statement of Work
Single-Photon Emission Computed Tomography
The Joint Commission

Section 3: Government Furnished Property, Equipment, and Services

None

Section 4: Contractor Furnished Items and Services

The Contractor shall provide all equipment, supplies, management, supervision, personnel, and transportation necessary to assure that all services are in accordance with the contract and all applicable laws and regulations. The contractor shall ensure all work meets performance standards specified in this Statement of Work (SOW).

Section 5: Specific Tasks

The Contractor shall comply with radiation protection standards in 29 CFR 1910.1096 and immediately report any unsafe conditions with the potential to adversely impact the facility radiation workers or patients to the Radiation Safety Officer (RSO).

5.1 General Requirements

- A.** All Radiation Physicists performing services under this contract shall work under a Qualified Medical Physicist (QMP). A Qualified Medical Physicist is a person who is certified by the American Board of Radiology, American Board of Medical Physics, or the Canadian College of Physicists in Medicine. The contractors must submit evidence of board certification with their quote and solicitation submittal.
- B.** All Radiation Physicists performing services under this contract shall ensure strict compliance to the release of protected Health Information in accordance with the Code of Federal Regulation, Title 45, Public Welfare, Subtitle A, Department of Health and Human Services, Part 164, Security and Privacy (45 CFR 164) and AFI 41-210, Patient Administration Functions, Health Insurance Portability and Accountability Act (HIPPA) and, General Guidelines on Releasing Medical Information.

5.2 Required Services

5.3

A. Equipment Inspections

- A.** All imaging equipment shall be inspected at least annually, not to exceed 14 months. Other inspections as defined below (new equipment, repairs, shielding (lead aprons), etc. shall be completed as needed. A complete electronic report in “pdf” format shall be provided to the Radiology Supervisor and Biomedical Engineering or a printed (hard copy) version of the report results shall be provided within 30 days after completion of the inspection.
- B.** Missed or errors in performance for required elements will be corrected at the expense of the contractor. If errors are found in reports, corrections in “pdf” form will be submitted expeditiously.

- C. The qualified diagnostic Medical Physicist shall perform imaging equipment (X-ray equipment, Mammogram Unit, Ultrasound units, CT, and Dental units) inspections to ensure compliance with the current Regulatory agency requirements. Any deficiencies or non-conformances discovered during the inspection shall be verbally communicated to Biomedical Engineering and the RSO prior to the qualified diagnostic medical physicist leaving the facility. Deficiencies or non-conformances which represent unsafe conditions with the potential to adversely impact the facility radiation workers or patients must be reported to the RSO immediately upon discovery.
- D. The qualified diagnostic medical physicist shall perform acceptance testing of all new or relocated imaging equipment upon installation and prior to first clinical use. The contractor shall verify that any new Radiology equipment meets manufacturer specifications and regulatory requirements, as well as identify any problem areas that need to be addressed. The acceptance testing shall comply with ACR and MQSA requirements. Any deficiencies or non-conformances discovered during the inspection shall be verbally communicated to the Radiology Supervisor, Biomedical Engineering and RSO prior to the qualified diagnostic medical physicist leaving the facility. Deficiencies or non-conformances which represent unsafe conditions with the potential to adversely impact the facility radiation workers or patients shall be reported to Biomedical Engineering and the RSO immediately upon discovery.
- E. The qualified diagnostic medical physicist shall perform a full inspection of imaging equipment after repairs or modifications that may affect the radiation output or image quality where required. In cases where the unit may not be used clinically until inspected by a physicist, the inspection must be completed within 72 hours after the facility contacts the contractor. Any deficiencies or non-conformances discovered during the inspection shall be verbally communicated to the Radiology Supervisor, Biomedical Engineering and RSO prior to the qualified diagnostic medical physicist leaving the facility. Deficiencies or non-conformances which represent unsafe conditions with the potential to adversely impact the facility radiation workers or patients shall be reported to the Radiology Supervisor, Biomedical Engineering and RSO immediately upon discovery. Where acceptable, the physicist will review and accept the repairs report from Engineering.

- F.** The qualified diagnostic medical physicist shall provide consultation for additional services as needed, such as safety training, reading dosimetry badge reports, i.e.
- G.** The qualified diagnostic medical physicist shall review CT protocols at least annually.

B. Radiation Safety Audit

- H.** The qualified diagnostic medical physicist will review the program and perform a Radiation Safety Audit for the Rosebud Service Unit.
- I.** The qualified diagnostic medical physicist shall provide shielding design calculations for each new, replaced, or relocated x-ray imaging system. The calculations for each shall comply with the National Council for Radiation Protection and Measurements (NCRP) Report No. 147, and, for dental units, NCRP Report No. 145, and shall be documented in a written report which includes a diagram showing adjacent areas. The qualified diagnostic medical physicist shall perform a shielding survey to verify the structural shielding was installed per the shielding design report and complies with the design goals.
- J.** The qualified diagnostic medical physicist shall assist in the development of a comprehensive technical quality assurance (QA) program (e.g., technique charts, repeat/reject analysis monitoring, monitoring of exposure indices to radiographic image receptors, QA program for display monitors, QA for CT, monitoring of dose metrics from fluoroscopy studies), which complies with ACR recommendations, for all modalities. The qualified diagnostic medical physicist shall review, at least annually, the QA program

- K. The qualified diagnostic medical physicist shall review the report from Engineering to verify compliance of any necessary corrective action performed to correct deficiencies found. The contractor will perform a follow-up inspection if necessary according to ACR/MQSA or other relevant standards.

5.4 Equipment Inspections Descriptions

The Contractor shall conduct equipment inspections or quality control surveys of the imaging equipment listed below. The Contractor shall ensure the imaging equipment's compliance with applicable Federal regulations and ACR recommendations and shall include, but not be limited to, monitoring the following basic performance characteristics.

A. Radiographic and Fluoroscopic Equipment

Physics inspections of radiographic and fluoroscopic equipment shall comply with the ACR Technical Standard for Diagnostic Medical Physics Performance Monitoring of Radiographic and Fluoroscopic Equipment. The performance of each radiographic and fluoroscopic unit must be evaluated at least annually. This evaluation should include, but not be limited to, the following tests (as applicable):

- 1) Integrity of unit assembly.
- 2) Collimation and radiation beam alignment.
- 3) Fluoroscopic system spatial resolution.
- 4) Automatic exposure control system performance.
- 5) Fluoroscopic automatic brightness control performance (high-dose-rate, pulsed modes, field- of-view [FOV] variation).
- 6) Image artifacts.
- 7) Fluoroscopic phantom image quality.
- 8) kVp accuracy and reproducibility.
- 9) Linearity of exposure versus mA or mAs.
- 10) Exposure reproducibility.
- 11) Timer accuracy.
- 12) Beam quality assessment (half-value layer).
- 13) Fluoroscopic entrance exposure rate (or air kerma rate). Maximum output and output using a phantom representing a standard size patient for all clinically used settings. [The mode of operation [e.g., magnification mode, frame rate, and any other mode selected] must be documented for each measurement.]
- 14) Fluorographic (image recording) entrance exposure rate (or air kerma rate) for cine imaging, if performed and entrance exposure (or air kerma) for spot images (if

performed). Maximum output and output using a phantom representing a standard size patient for all clinically used settings. [The mode of operation (e.g., magnification mode, frame rate, etc.) must be documented for each measurement.]

- 15) Image receptor entrance exposure.
- 16) Equipment radiation safety functions.
- 17) Patient dose monitoring system calibration. This includes, for radiographic systems, the metric of dose to the image receptor (IEC Exposure Index or proprietary index) and, for fluoroscopy systems, the displays of cumulative air kerma and, if available, DAP.
- 18) Display monitor performance.
- 19) Digital image receptor performance.
- 20) Grids used with portable x-ray units shall be imaged for uniformity.
- 21) For radiographic units, measurement of entrance skin exposure (or air kerma) for a standard size patient for common radiographic projections and comparison to published diagnostic reference levels and achievable doses (e.g., ACR practice parameter).

Note: The information on entrance exposure rates (or air kerma rates) from fluoroscopy and from fluorography, in Items (13) and (14) above, for each fluoroscope, shall be in a format suitable for providing to the physicians who operate the fluoroscope.

B. Digital Radiography (DR)

Physics inspections of DR equipment shall comply with the American Association of Physicist in Medicine (AAPM) Report Number 93, Acceptance Testing and Quality Control of Photostimulable Storage Phosphor Imaging Systems. The performance of DR must be evaluated at least annually. This evaluation should include, but not be limited to, the following tests (as applicable):

- 1) Component and Imaging Plate Physical Inspection and Inventory.
- 2) Imaging Plate Dark Noise and Uniformity.
- 3) Exposure Indicator Calibration.
- 4) Linearity and Auto-ranging Response.
- 5) Laser Beam Function.
- 6) Limiting Resolution and Resolution Uniformity.
- 7) Noise and Low-Contrast Resolution.
- 8) Spatial Accuracy.
- 9) Erasure Thoroughness.
- 10) Aliasing/Grid Response.
- 11) IP Throughput.
- 12) Positioning and Collimation Errors.

C. CT Scanners

The physics inspection shall conform to the 2012 ACR Computed Tomography Quality Control Manual or latest revision. The performance of each CT scanner shall be evaluated at least annually. This evaluation should include, but not be limited to, the following tests (as applicable):

Review of Clinical Protocols.

- 1) Scout Prescription and Alignment Light Accuracy.
- 2) Image Thickness – Axial Mode.
- 3) Table Travel Accuracy.
- 4) Radiation Beam Width.
- 5) Low-Contrast Performance.
- 6) Spatial Resolution.
- 7) CT Number Accuracy.
- 8) Artifact Evaluation.
- 9) CT Number Uniformity.
- 10) Dosimetry (the scanner displayed CTDIvol values must be within +/- 20% of the measured CTDIvol values).
- 11) Gray Level Performance of CT Acquisition Display Monitors.

D. Dental

The physics inspection shall conform to the Conference of Radiation Control Program Directors (CRCPD), Quality Control Recommendations for Diagnostic Radiography Volume 1 Dental Facilities July 2001. The performance of dental x-ray inspections shall be annually. This evaluation should include, but not be limited to, the following tests (as applicable).

- 1) Collimation.
- 2) Beam quality (half value layer).
- 3) Timer Accuracy and Reproducibility.
- 4) kVp Accuracy and Reproducibility.
- 5) mA or mAs Linearity.
- 6) Exposure Reproducibility.
- 7) Entrance Skin Exposure Evaluation, with comparison to published diagnostic reference levels and achievable doses (e.g., NCRP Report No. 172).
- 8) Technique Chart Evaluation.
- 9) Image uniformity (artifact evaluation).

Dental Cone Beam Computed Tomography (CBCT) Acceptance and Performance Testing

- a) **Acceptance Testing.** Acceptance testing and measurements of air kerma at the isocenter for each kVp station for a range of clinically used mAs settings will be performed initially when the CBCT unit is installed, and following any move of the CBCT to another area inside or outside the initial clinical site. This testing is to ensure that the equipment

performance is in agreement with the manufacturer's technical specifications.

- b) **Performance Testing.** Each CBCT unit shall undergo periodic quality control tests to ensure that the performance of the machine has not significantly deteriorated and is operating within the manufacturer's technical specifications. This performance testing is performed by a qualified expert annually, at intervals not to exceed 14 months, and after repairs to the CBCT unit that may affect the radiation output or image quality.
- c) Some manufacturers provide a phantom and procedures to perform machine specific quality assurance (QA) tests. In cases where the manufacturer provides a phantom and procedures to perform specific tests but the tests are not included in this SOW, then the manufacturer's machine-specific QA tests shall be performed in addition to the QA tests in this SOW.

Acceptance and Annual physics testing for Dental CBCT

a) Radiation output Repeatability

Make four measurements of the air kerma at the isocenter at a clinically used setting. The measurements should be less than $\pm 5\%$ of the average of the five measurements and the measurements should be less than $\pm 5\%$ of the previous year's measurement.

b) Radiation Output Reproducibility

Measure the air kerma at the isocenter for each kVp station and a range of clinically used mAs setting. Compare the results to the baseline values established at the initial acceptance testing. The values should be $\pm 5\%$ of the baseline.

c) kVp Accuracy

Measure the kVp at all clinically used settings. The measured kVp should be $\pm 5\%$ of the selected kVp.

d) kVp Repeatability

Make five kVp measurements each for two clinically used kVp settings. All measured values should be $\pm 5\%$ of the mean kVp.

e) kVp Reproducibility

Measure the kVp at all available kVp settings. The measured values should be $\pm 5\%$ of the baseline.

f) Beam quality

Measure the half value layer (HVL) for aluminum. The minimum shall comply with Section F.4.d of the Suggested State Regulations for Control of Radiation, Conference of Radiation Control Program Directors.

g) Radiation field of view (FOV)

Measure the width of the radiation field at the isocenter. The width of the beam should be 3 mm or 30% of the total nominal collimated width.

h) Image Quality

Image the phantom provided by the manufacturer or another suitable phantom. Assess high contrast spatial resolution, uniformity of trans axial images, and image noise. Imaging uniformity shall be assessed over the entire range of axial images.

i) Accuracy of Linear Measurements

Using images of an appropriate phantom, assess the accuracy of distance measurements.

j) Accuracy of Patient Dose Metric Indication

Assess the accuracy of the indicated dose metric (typically DAP).

k) Patient Dose Assessment

From a scan or scans using the facility's standard techniques, record the dose metric (typically DAP) and compare to achievable levels and diagnostic reference levels (if available)

l) Review of the technical QA program

The qualified expert shall review the technical QA program. The review shall include a trend analysis of the QA data. The results of the technical QA program review shall be included in the written report. Any trends that identify problems shall be included in the report along with recommended corrective actions.

m) Display Monitors

Perform a visual analysis of the SMPTE test pattern.

- i. Display the test pattern on the imaging console. Set the display window width/level to the manufacturer-specified values for the pattern. Do not set the window/level by eye; doing so invalidates the procedure.
- ii. Examine the pattern to confirm that the gray level display in the imaging console

is subjectively correct.

1. Review the line pair patterns in the center and at each of the corners.
 2. Review the black-white transition.
 3. Look for any evidence of “scalloping” (loss of bit depth) or geometric distortion.
- iii. Use a photometer to measure the maximum and minimum monitor brightness (0% and 100% steps)
- iv. Measure additional steps within the pattern to establish a response curve.
- v. Measure the brightness near the center of the monitor and near all 4 corners (or all 4 sides, depending on the test pattern used).
- n) Viewing Conditions
Assess the viewing conditions for the area in which the monitor used to evaluation the CBCT studies is located.

E. ULTRASOUND

The physics inspection shall conform to the ACR performance tests for ultrasound. Testing should be done for all transducers clinically used with any unit employing more than one transducer. Ultrasound equipment shall be inspected semi-annually (every 6 months).

1.) System Sensitivity/ Penetration

This test should be done with the following settings:

- Maximum transmit power
- Proper receiver gain and TGC that allows the echo texture to be visible in the deep region
- Transmit focus at the deepest depth

The maximum depth of visualization is determined by comparing the gradually weakening echo texture to electronic noises near the bottom of the image.

2.) Image Uniformity

Adjust the TGC controls and other sensitivity controls to obtain an image as uniform as possible.

- Vertical or radially oriented streaks?
- Dropouts?
- Reduction of brightness near edges of the scan?
- Brightness transitions between focal zones?

3.) Electrical and Mechanical Safety and Cleanliness

- Are all cords and cables intact (no frays)?
- Are all transducers intact without cracks or delamination?
- Are the transducers cleaned after each use?
- Are the image monitors clean?
- Are the air filters clean?
- Are the wheel locks in working condition?
- Are the wheels fastened securely to the US unit and so the wheels rotate easily?
- Are all accessories (VCR, cameras, etc.) fastened securely to the US unit?

4.) Gray Scale Photography (if applicable) – Do either (a), (b) or (c).

a.) For Scanners with a Discrete Bar Pattern

Count the number of distinct gray bar steps on the viewing monitor. Then count the number of steps visualized in the gray bar on the hard copy image.

b.) For Scanners with a Continuous Gray Bar Pattern

Use calipers to measure the length of the black-to-white transition of the gray wedge on the viewing monitor. If the relative length of the black-to-white transition on the hard copy image is less, document how much is missing.

c.) For Laser Imager (Hard Copy Device)

Prior to filming any images, an SMPTE test pattern created by the Society of Motion Picture and Television Engineers (SMPTE), should be printed using the appropriate window width (WW) and window level (WL). If you are unfamiliar with this procedure, you should review Gray et al., “ Test pattern for video display and hard-copy camera,” Radiology 145:519-527 (1985), and then contact your local service engineer for assistance. When printed, the 95% density patch within the 100% square and the 5% density patch within the 0% square should be visible, and there should be no notable distortions or artifacts present. If these criteria are not met, contact your service engineer for laser camera before proceeding with any filming.

5.) Hard Copy Output Quality Test (Digital)(if applicable)

This test, or a similar test specifically recommended by the hard copy equipment manufacturer.

Required Test Equipment

- Densitometer
- SMPTE Test Pattern or another similar test pattern or phantom image having a wide range of gray scales.

The same test image should be used each time.

F.) Display Monitors

The annual performance evaluation conducted by the diagnostic medical physicist includes testing of image acquisition display monitors for maximum and minimum luminance, luminance uniformity, resolution, and spatial accuracy. The image acquisition display monitors for CT and Mammogram units shall be tested.

G.) Mammography-MQSA testing per requirement

The qualified diagnostic medical physicist inspecting mammography equipment must meet the qualifications outlined in the MQSA and shall provide the facility with up to date documentation demonstrating the qualified diagnostic medical physicist is MQSA qualified. Inspections of mammography equipment must comply with the latest requirement posted on the ACR Web site for the manufacturer of the digital mammography unit being inspected. Inspection items may include:

- Mammographic Unit Assembly Evaluation
- Collimation Assessment
- Artifact Evaluation
- kVp accuracy and reproducibility
- Beam quality assessment-HVL measurements
- Evaluation of system resolution
- Automatic Exposure Control (AEC) function performance
- Breast entrance exposure, AEC reproducibility, and average glandular dose
- Radiation output rate
- Phantom image quality evaluation
- Signal-to-noise ratio and contrast-to-noise ratio measurements
- View box luminance and room illuminance
- Review Workstation (RWS) tests

5.5 Completion of Service and Documentation: On completion of all performance monitoring and testing services, the contractor's representative shall brief the Radiology Supervisor on all aspects pertaining to the Radiology Department. The Dental Director shall also be briefed on all aspects pertaining to the Dental Department. Documentation of all services that were performed shall be provided; in addition, a copy of the report is to be provided to the Rosebud Service unit Radiology Supervisor and Clinical Director.

5.6 Equipment List

Description	X-ray Units	Planned replacement or addition
ROSEBUD X-RAY UNITS		
CT: Siemens SOMATOM 128 slice	1	
Fluoro & X-ray: GE Discovery RF 180	1	
Quantum DR Radiographic Room	1	
Schimidzu/ MobileDart Portables	2	
Canon Aplio i700 Ultrasound	1	
Canon Aplio MX Ultrasound	1	
Hologic Selenia Dimension 500 Mammo Unit	1	
Lead Garments (Aprons)	50	
ROSEBUD DENTAL		
Progeny Dental Bitewing Preva	8	
Aribex Nomad Pro 2 hand held bitewing	1	
Sirona Orthophos S	1	